

Physical vs Chemical Changes

Year 6 Science | Australian Curriculum v9.0 | AC9S6U04 | Reversible & irreversible changes

Name: _____ Date: _____

LEARNING INTENTION & SUCCESS CRITERIA

I can compare changes using evidence about new substances.

I can explain physical and chemical changes and describe familiar reversible and irreversible examples.

KEY CONCEPTS

Physical change: a change in form, size or state without forming a new substance. Examples: ice melting, paper tearing, salt dissolving.

Chemical change: new substances form. Examples: iron rusting and an egg cooking.

Important: being difficult to undo does not, by itself, prove a chemical change.

WORKED EXAMPLE

Ice melts: it is still water, so the change is physical. Freezing can reverse it. Iron rusts: rust is a new substance, so this is a chemical change and cannot be reversed by simply cooling or reshaping.

PART A - CLASSIFY & EXPLAIN

Write physical or chemical. Give a reason about the substance.

1. Water freezes into ice.

2. A steel bike chain develops rust.

3. A sheet of paper is cut into strips.

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PART B - CORE SKILLS

4. Compare melting butter with cooking an egg. Which is a physical change and which is a chemical change? Explain.

5. Salt dissolves in water. Explain why this is a physical change and how the salt can be recovered.

6. Water boils and makes bubbles. Do bubbles alone prove a chemical change? Explain.

PART C - CHALLENGE

7. Noah says, "Tearing paper is chemical because I cannot turn the pieces back into the original sheet." Evaluate his claim using the idea of a new substance.

REFLECTION & I CAN CHECK

The best question to ask when classifying a change is:

I can explain a change using evidence: Yes / Not yet

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ANSWER KEY

1. Physical: water changes from liquid to solid; it remains water. Melting reverses it.
2. Chemical: rust forms as iron reacts with oxygen and water. Rust differs from the original iron.
3. Physical: the paper changes size/shape but remains paper.
4. Simply melting butter is physical: it changes form and can solidify on cooling. Cooking an egg causes chemical changes; cooling does not restore the raw egg.
5. Salt remains present in the solution; no new substance forms in this example. Evaporating the water recovers salt.
6. No. Boiling makes water-vapour bubbles. Water changes state without forming a new substance.
7. Noah's claim is incorrect. Difficulty reversing the change is not sufficient evidence of a chemical change. Tearing changes the paper's shape/size; the material remains paper.

Reflection: "Has a new substance formed?" Accept similar wording.

TEACHING & PARENT TIP

Use photographs of rusting and cooking, and demonstrate ice melting or paper cutting. Ask learners to separate an observation ("the colour changed") from an explanation ("a new substance formed").

Colour change, gas production or temperature change can be clues, but require context. Avoid teaching "all physical changes are reversible" as a rule. Adult supervision is required for any heating demonstration.

TEACHER FEEDBACK

Strength: _____

Next step: _____